Decision Trees I

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1. The Regression Tree
2. Tree Growing
3. Tree Pruning
4. Building a Regression Tree

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How decision tree works?

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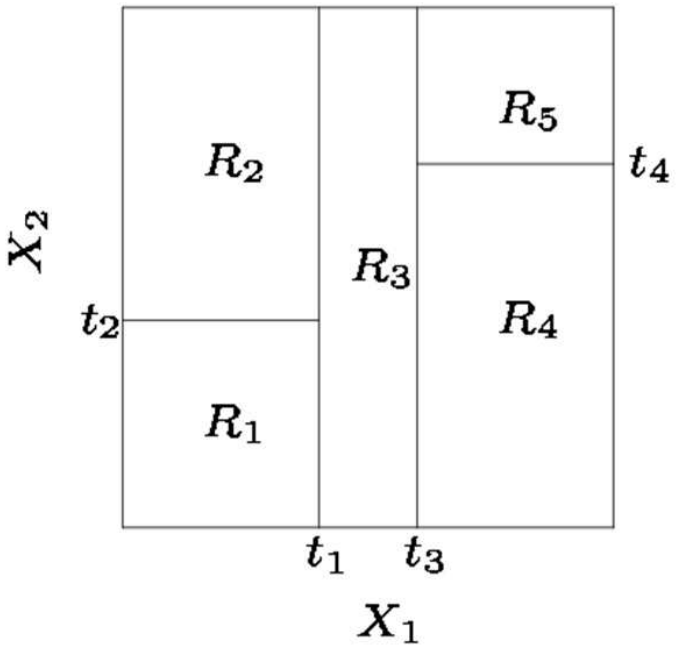
Predictor Space Partitioning

1. divide the predictor space (i.e. all the possible values for for $X_1, X_2, ..., X_p$) into distinct regions, say $R_1, R_2, ..., R_k$.
2. for every x_i that falls in a particular region (say R_j) we make the same prediction.



How decision tree works?

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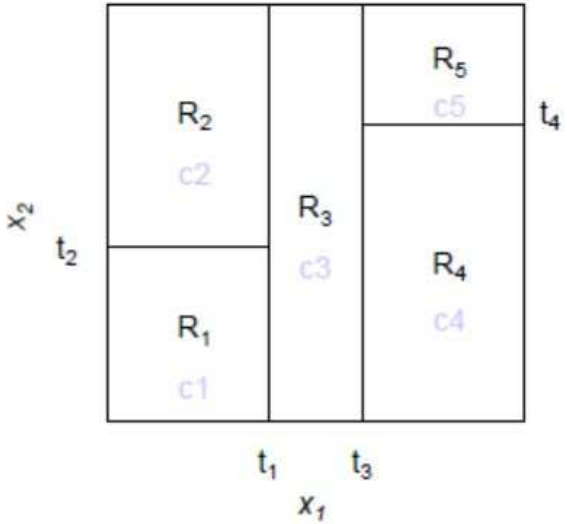
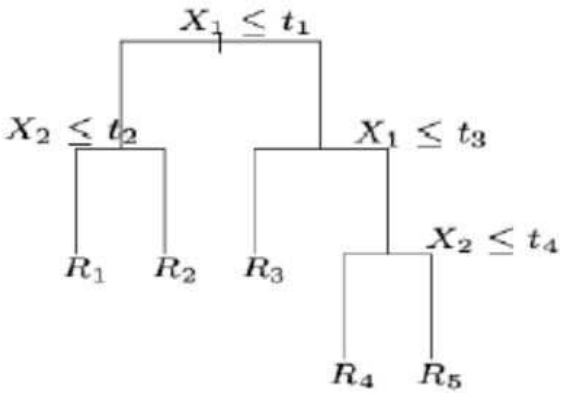
Predicting the response

1. Suppose that c_j is the regression model prediction value corresponding to the region R_j.
2. $\hat{c}_j = average(y_i | x_i \in R_j)$.
3. $\hat{y}_i = f(x_i) = \sum_{j=1}^{k} \hat{c}_j I(x_i \in R_j)$





An Example

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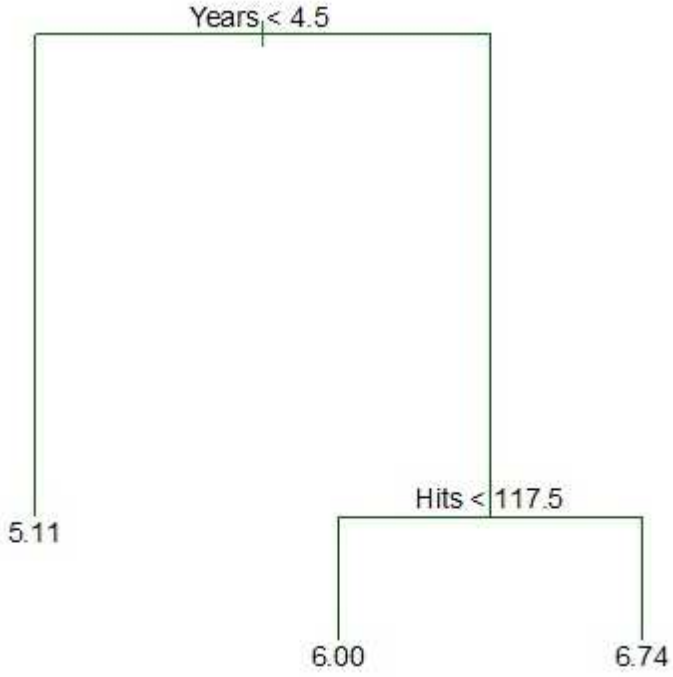
Predicting Baseball Players' Salaries

Try to predict the salary for a player who played in the league for 5 years and had 115 hits last year. (Salary is measured in 1000s, and log-transformed)



Predicting Baseball Players' Salaries

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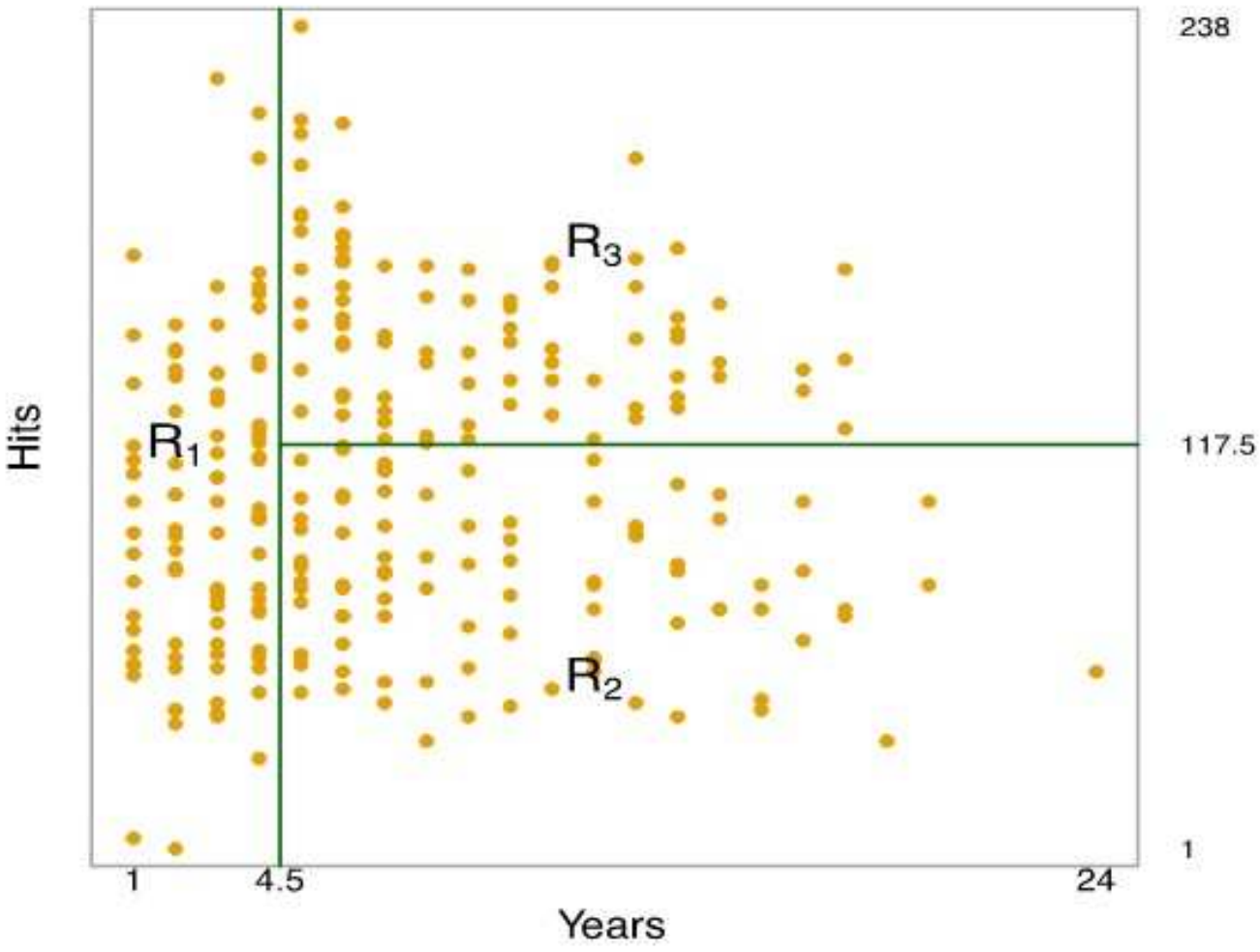

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How to construct the regions

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The objective function

The goal is to find regions $R_1, ..., R_k$ that minimize the RSS given by,

$$\sum_{j=1}^{k} \sum_{i \in R_j} (y_i - \hat{y_{R_j}})^2 \tag{2.1}$$

where $\hat{y_{R_j}}$ is the mean response for the training observations within the jth region.

Solution

A top-down, greedy approach that is known as recursive binary splitting.

How to construct the regions

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Where to Split

Given the predictor X_j and the cutpoint s, for any j and s, the predictor space is split into the following two regions

$$R_1(j,s) = \{X|X_j < s\}, R_2(j,s) = \{X|X_j \geq s\}, \tag{2.2}$$

and we seek the value of j and s that minimize the equation

$$\sum_{i:x_i \in R_1(j,s)} (y_i - \hat{y_{R_1}})^2 + \sum_{i:x_i \in R_2(j,s)} (y_i - \hat{y_{R_2}})^2 \tag{2.3}$$

When to Stop

Until our regions have too few observations to continue.

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Cost complexity pruning

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Cost complexity pruning

Rather than considering every possible subtree, we consider a sequence of trees indexed by a nonnegative tuning parameter α. For each value of α there corresponds a subtree $T \subset T_0$ such that

$$\sum_{j=1}^{|T|} \sum_{i:x_i \in R_j} (y_i - \hat{y_{R_j}})^2 + \alpha |T| \tag{3.1}$$

is as small as possible. Here $|T|$ indicates the number of terminal nodes of the tree T.

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Algorithm 8.1

1. Use recursive binary splitting to grow a large tree on the training data, stopping only when each terminal node has fewer than some minimum number of observations.
2. Apply cost complexity pruning to the large tree in order to obtain α sequence of best subtrees, as a function of α.
3. Use K-fold cross-validation to choose α. For each $k = 1, ..., K$:
   1. Repeat Steps 1 and 2 on the $\frac{K-1}{K}$th fraction of the training data, excluding the kth fold.
   2. Evaluate the mean squared prediction error on the data in the left-out kth fold, as a function of α.
4. Return the subtree from Step 2 that corresponds to the chosen value of α.

Questions and Answers

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Thank you for your attention!